



Name:- Jaykishan J Saharan

Div:-D Roll no:-26

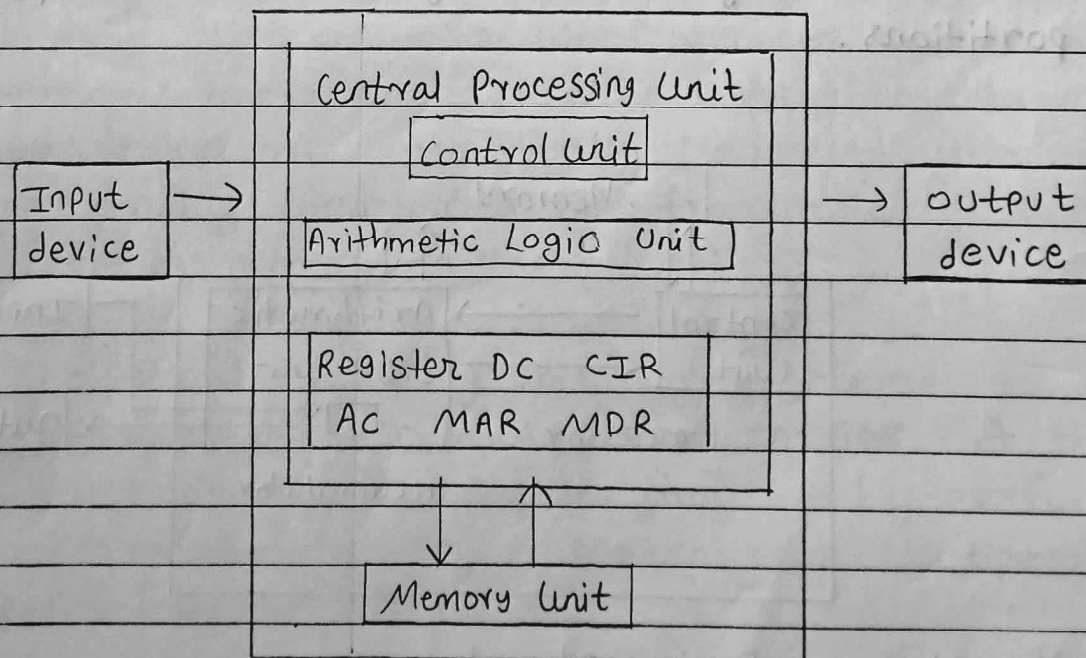
Course/Program:- BTech CSE 8Y

Subject:- computer organization & Architecture.

COA Theory Assignment 2

Q.1) Explain Von Neuman Architecture with diagram? also explain its key features?

Sol:- Von neumann architecture was first published by John von Neumann 19 1945. His computer architecture design consists of a control unit, Arithmetic and logic unit (ALU), Memory unit, Register and Input outputs. Von Neumann Architecture is based on the stored program computer concept, where instruction data and program data are stored in the same memory. This design is still used in most computer produced nowadays.



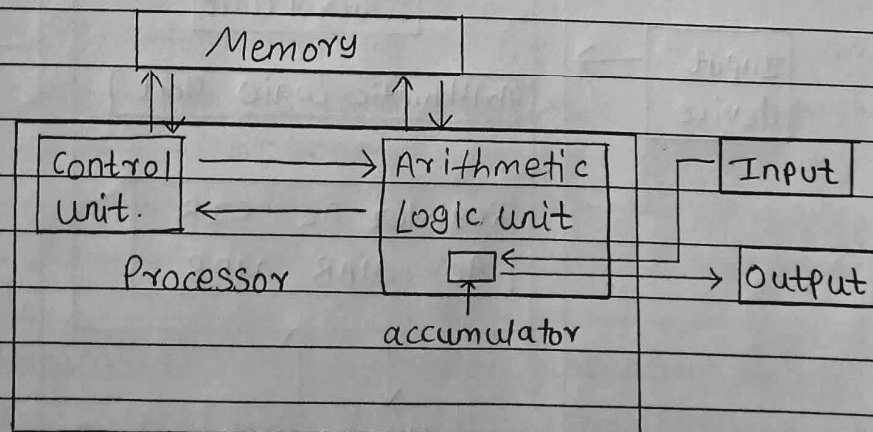
Key features:

ALU :- Perform arithmetic (add, subtract) and Logic (AND, OR, NOT) operations, to be carried out.

CU :- control unit controls the operation of computer's ALU, memory and input/output devices, telling them how to respond to program instructions it has just read and interpreted from the memory unit. The CU also provides the timing and control signals required by other computer components.

Register: Register are high speed storage areas in the CPU. all data must be stored in register before it can be processed. eg: MDR, AC, DC, CYR.

Memory unit: The memory unit consists of RAM, sometimes referred to as primary or main memory. It is fast and also directly accessible by CPU. RAM is split into partitions.

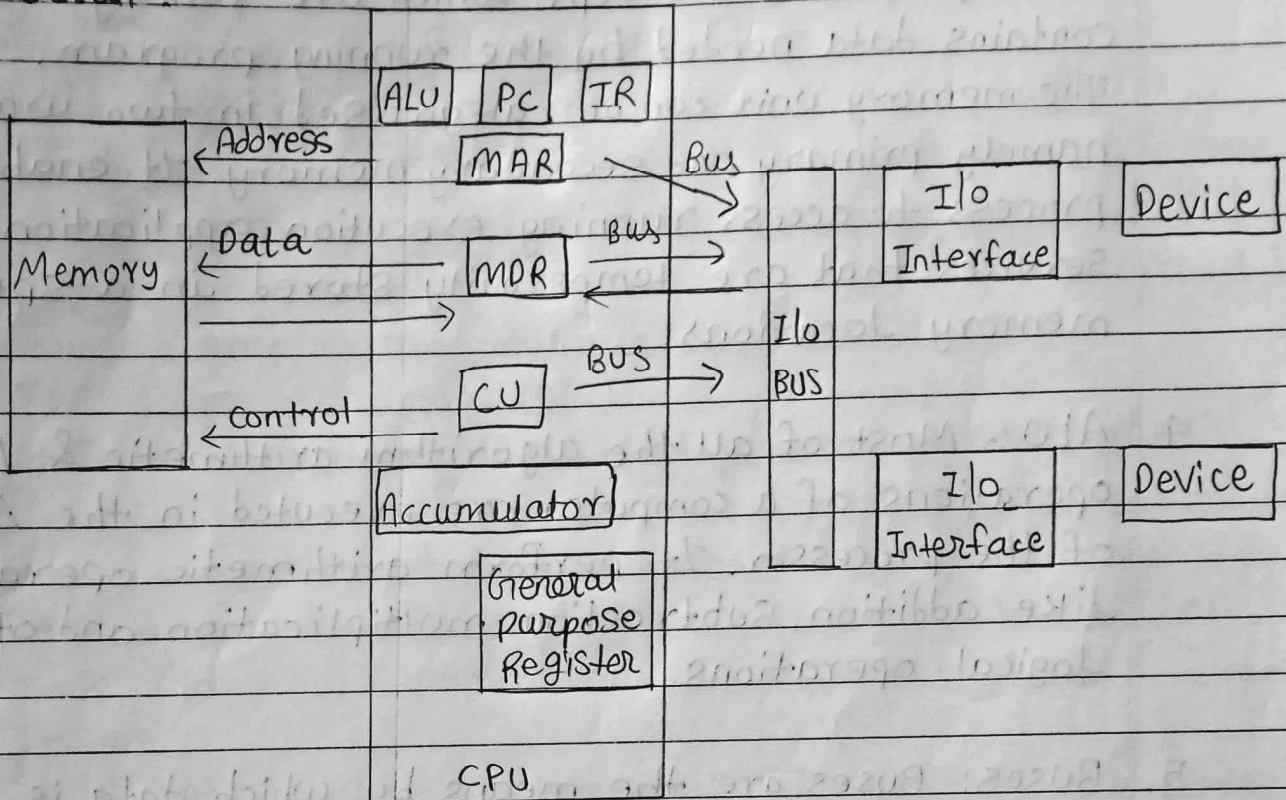


Von Neuman Basic structure



Q.2) Draw diagram of Basic Structure of CPU & explain in detail?

Solⁿ:-



1. **Input unit**:- Input units are used by the computer to read the data. Most commonly used inputs devices are mouse, keyboards, microphones etc. The instructions through these devices are automatically translated into its corresponding binary code and transmitted over a cable to either memory or the processor.
2. **CPU**:- Central Processing unit can be referred as electronic circuitry within a computer that carries out the instructions given by computer program by performing basic arithmetic, logic, control and I/O operations specified by the instructions.



3. **Memory Unit:** It can be referred as the storage area in which programs are kept which are running and that contains data needed by the running program. The memory unit can be categorised in two ways namely primary and secondary memory. It enables a process to access running execution application and services that are temporarily stored in a specific memory location.
4. **ALU:-** Most of all the algorithm arithmetic & logical operations of a computer are executed in the ALU of the processor. It performs arithmetic operations like addition, subtraction, multiplication and other logical operations.
5. **Buses:** Buses are the means by which data is transmitted from one part of a computer to another connecting all major internal components to the CPU and memory.
6. **Control unit:** It is a component of computer's CPU that coordinates the operation of the processor. It tells the computer's memory, arithmetic & logic unit and input & output devices how to respond to a program instruction, also called the nerve of a CPU.
7. **Output unit:-** The primary function of the output unit is to send the processed result to the user. Output devices display information in a way that the user can understand. These devices display information that has been held



are generated within a computer.

Q.3) History of computers? Explain in details.

Soln:- A generation of computer refer to the specific improvements in computer technology with time. In 1946, electronic pathways called circuits were developed to perform the counting. In each new generation, circuit become smaller and more advanced.

(1946-1959)
ci) 1st Gen computers: The first generation (1946-1959) computer slow, huge & expensive vacuum tubes were used as basics components of CPU and memory. These system depended on batch operating system and punch cards. Magnetic & paper tape we used as output & input devices.

eg: ENIAC (Electronic Numerical Integrator And computer)
EDVAC, UNIVAC, IBM-701, IBM-650.

(ii) 2nd Gen Computers: (1959-1965)
This was the era of Transistor computer. They uses Transistor which were cheaper, compact & consuming less power. It made them faster than 1st generation computer. Assembly languages & programming Language's like (COBOL, FORTRAN) were used in these computers.
eg: IBM-1620, IBM-7094, CDC-3600

(1966-1970) (1965-1970)
(iii) 3rd Gen computers: The third generation (1966-1970) used Integrated circuits (IC) instead of Transistor. A single IC can pack huge no. of transistor which increased the power of a computer and reduced the cost.



These generation computer used real-time time sharing and distributed OS.

Also High level programming language like FORTRAN-II to IV COBOL, PASCAL, ALGOL were used in generation.

eg:

IBM[®]360, Honeywell - 6000 series, PDP → } Personal
TDC-316 } Data processor }

(1971-1980)

(iv) 4th Gen computer: The fourth generation (1971-1980) used very large scaled integrated (VLSI) circuits, a chip containing millions of transistors and circuits. These made this generation computer more impactful, compact fast and affordable. Programming language like C, C++, DBASE were also used in this generation, eg: PDP-11, STAR-1000

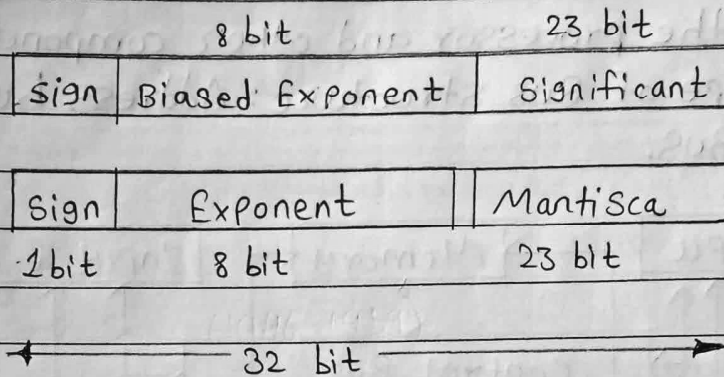
(v) 5th Gen computer [1980-till date]

Here VLSI is replaced by ultra large scale Integrated circuits. Programming let made possible the production of microprocessor chips with ten million electronic component. This generation computer used parallel processing Hardware and Artificial Intelligence (AI) Software. The programming Language used in the this generation were C, C++, Java, .NET etc.

eg: Desktop, laptop, Note-Book, chrome-Book.



Q.4) Explain IEEE standards for floating point arithmetic
Single precision-32 bit representation:- (IEEE 754)



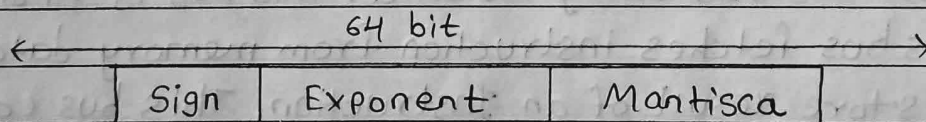
Theory point numbers are usually normalised.

Exponent is adjusted so that leading bit (MSB) of Mantissa is 1.

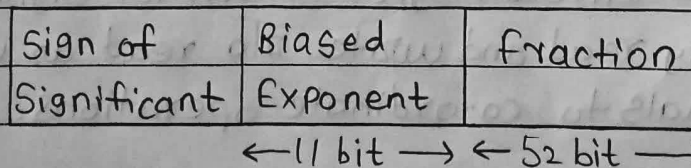
Since, it is always 1 there is no need to store it.

Scientific Notation where numbers are normalized to give a single digit before the decimal point like in decimal system eg: 3.123×10^3 .

Double Precision - 64 bit representation floating point numbers IEEE standard.

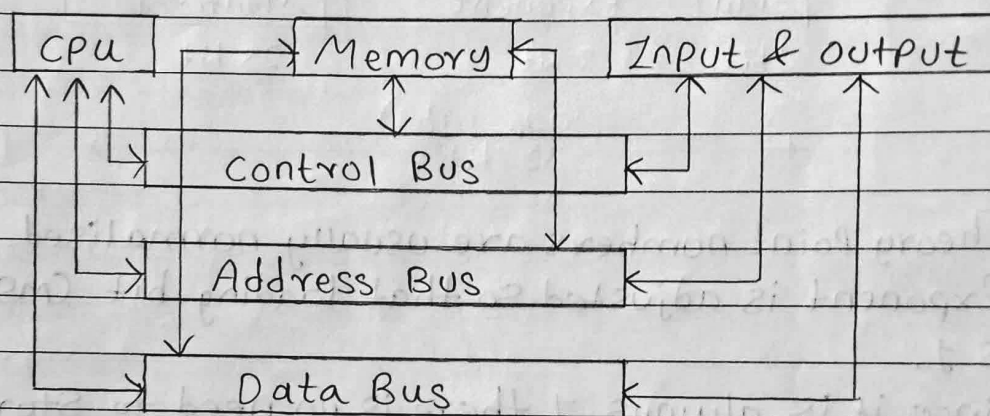


Double Precision: IEEE-754 floating point standard.



Q.5) Explain bus structure in detail.

Sol:- A Bus is collection of wires that connects several devices. Buses are used to send control signals and data between the processor and other components. There are 3 type of Bus structure: Address bus, Data Bus, control Bus.



Address Bus: It carry the memory address while reading from writing into memory. This bus carries input/output part address or device address from I/O part. Now a days computer having Bi-directional address bus.

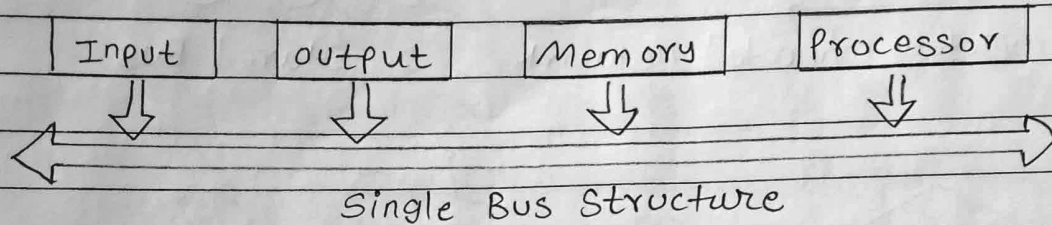
Data bus: This Bus carry the data. This bus is bi-directional bus. This bus fetches instruction from memory data Bus is used to store result of an Instruction. This bus carry command to an I/O devices.

control Bus:- Different types of control signals are used in a Bus. Memory read and write I/O read or write. It carries control signals to co-ordinate operations.



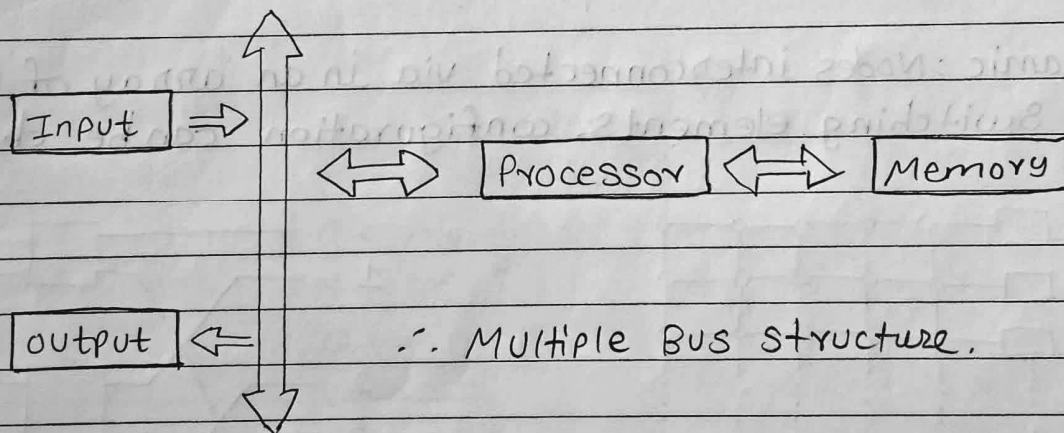
BUS Structure (Architectural Arrangement)

Single Bus Structure: one common bus for data, address and control.



Multiple Bus Structure:

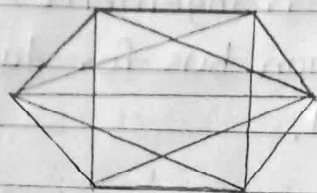
one bus is used to fetch instructions while other is used to fetch data, required for execution. It is to overcome the bottleneck of a single bus structure.



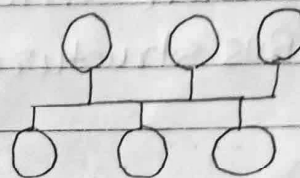
Q.6) Write in detail about interconnection network?

Solⁿ: Interconnection network are used to connect nodes where nodes can be a single processor or group of processor to the other nodes. It is of two types:-

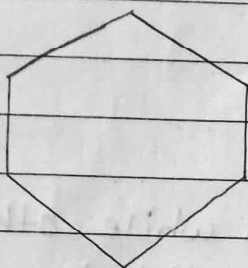
1. Static: These are hard wired and cannot change their configuration.



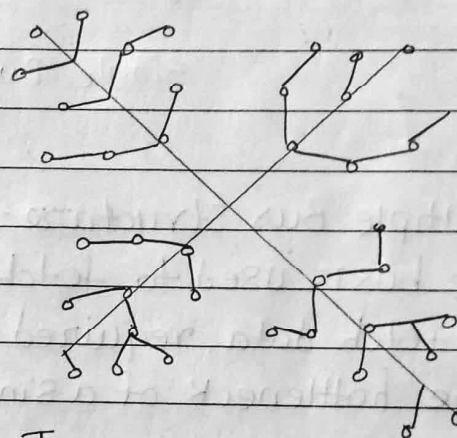
1. Completely connected network



2. Starred Bus

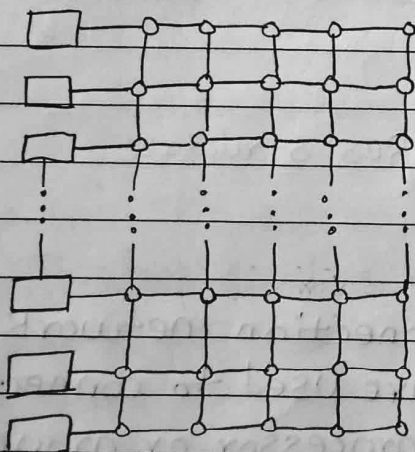


3. Ring

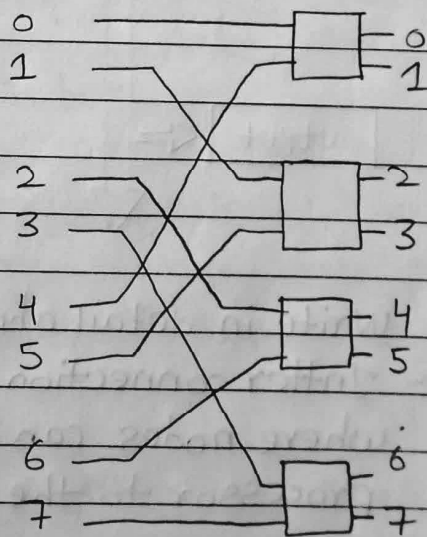


4. Tree

2. Dynamic : Nodes interconnected via in an array of simple Switching elements, configuration can be changed.



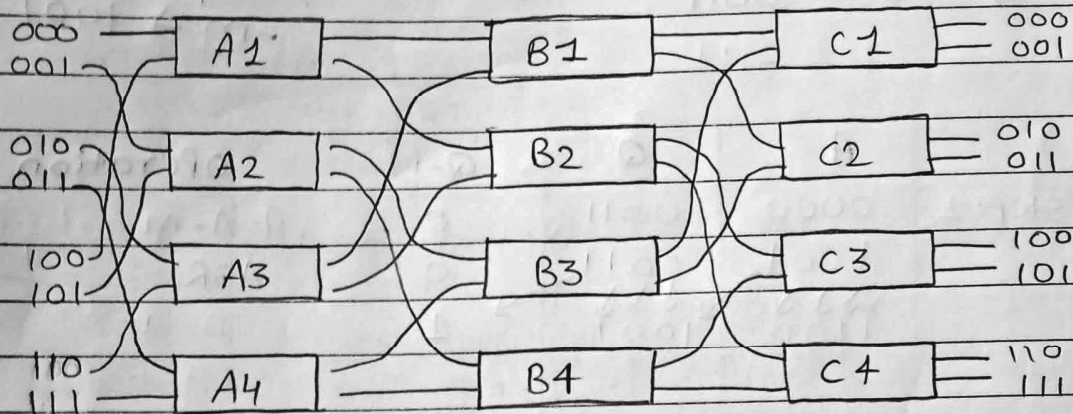
(cross-switch)



(single stage)

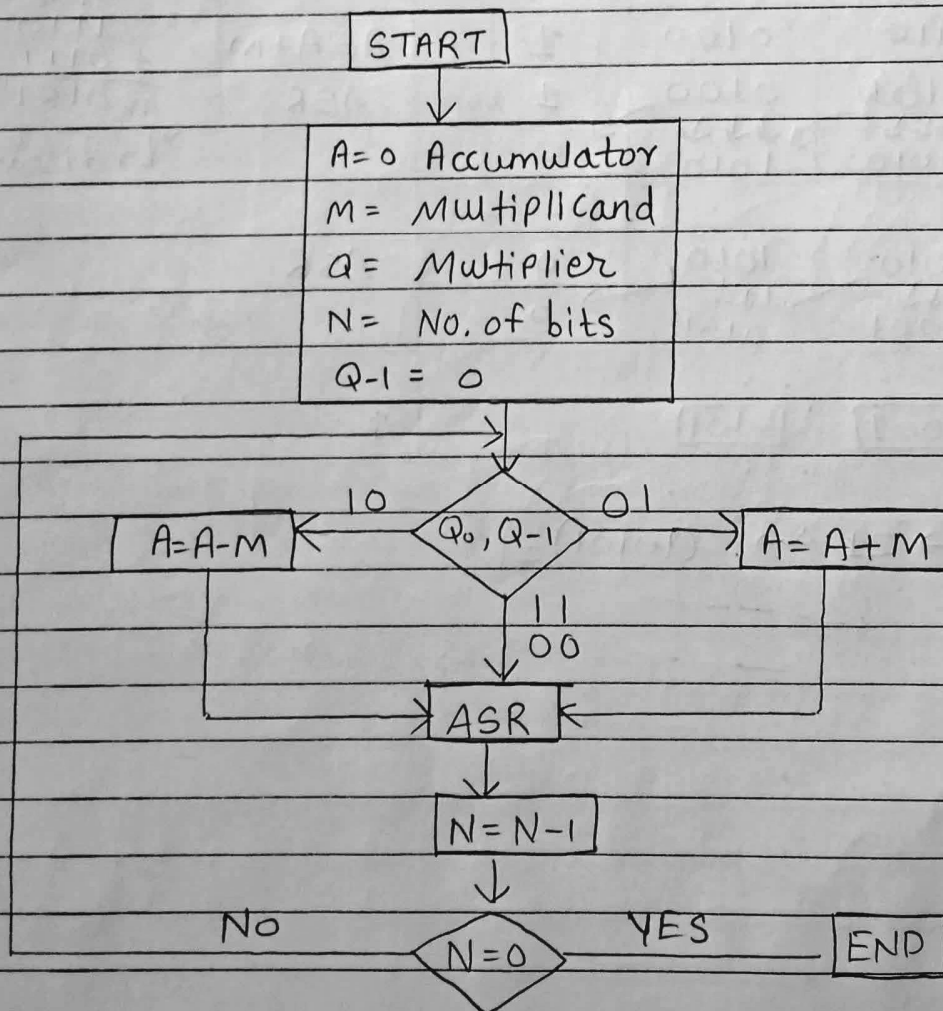


3. Multistage Interconnect Network:



Q.7) Explain Booth's Algorithm with flowchart? also Solve $7 * 3$ using Booth's algorithm.

Sol:-





Solve: $7 * 3 = ?$

$M = 37 \rightarrow 0111$

$Q = 73 \rightarrow 0011$

$-m: (1^s) \rightarrow 1000 (2^s)$

$+ \frac{1}{1001}$
 $-m \rightarrow 1001$

	A	Q	Q-1	operation
Step:1	0000	0011	0	$A = A - M$
	1001	0011	0	ASR
	1100	1001	1	
Step:2	1100	1001	1	ASR
	1110	0100	1	
	1110	0100	1	
Step:3	1110	0100	1	$A = A + M$
	0101	0100	1	ASR
	0010	1010	0	$\rightarrow$ discarded.
Step:4	0010	1010	0	ASR
	0001	0101	0	
	<u>0001</u>	<u>0101</u>		

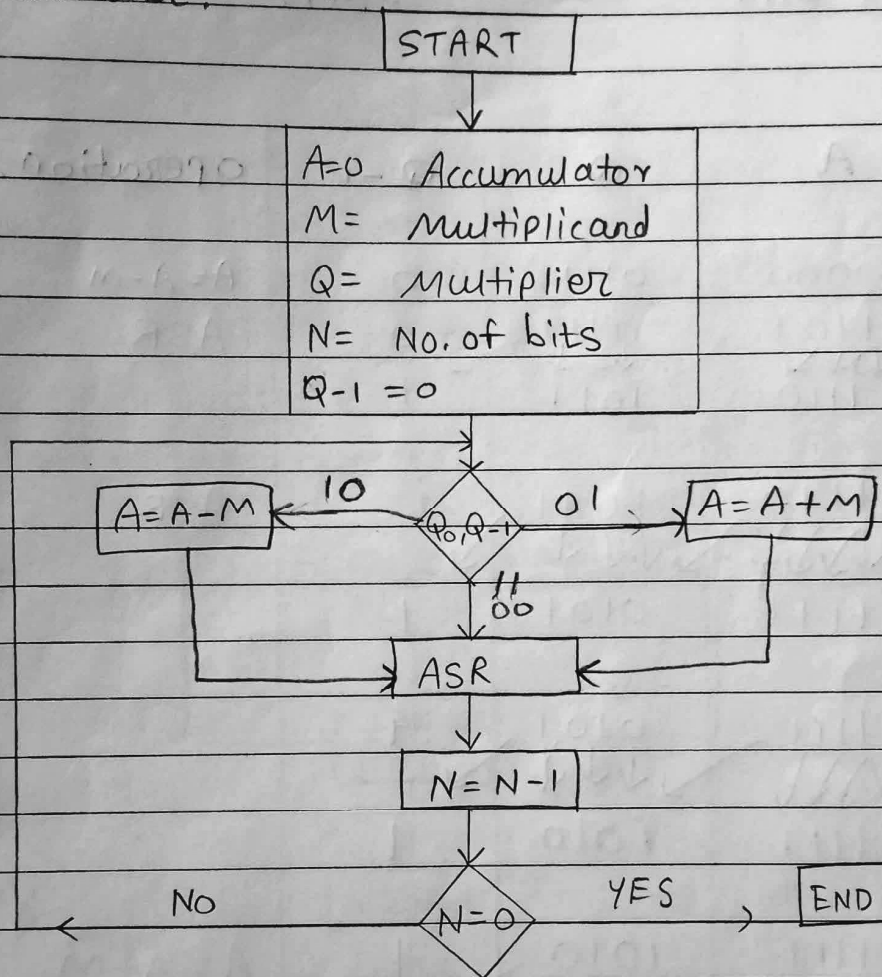
$3 \times 7 \rightarrow 21 \rightarrow (10101)_2$



Q.8) Solve $3 * 7$ using Booth's Algorithm? (For ALL)

Solⁿ:

flowchart





Solve: $3 \times 7 = ?$

$$m = 3 = 0011$$

$$Q = 7 = 0111$$

$$\begin{array}{r} -M \rightarrow 0011 \\ \rightarrow 1's \ 1100 \\ + \quad 1 \\ \hline -M \rightarrow 1's \rightarrow 1101 \end{array}$$

	A	Q	Q-1	operation.
Step:1	0000 1101 ↓ ↓ ↓ 1110	0111 0111 ↓ ↓ ↓ 1011	0 0 1	A = A - M ASR $\begin{array}{r} 0000 \\ + 1101 \\ \hline 1101 \end{array}$
Step:2	1110 ↓ ↓ ↓ 1111	1011 ↓ ↓ ↓ 0101	1 1	ASR
Step:3)	1111 ↓ ↓ ↓ 1111	0101 ↓ ↓ ↓ 1010	1 1	
Step:4)	1111 0101 ↓ ↓ ↓ 0001	1010 11010 ↓ ↓ ↓ 0101	1 1 0	A = A + M ASR ASR

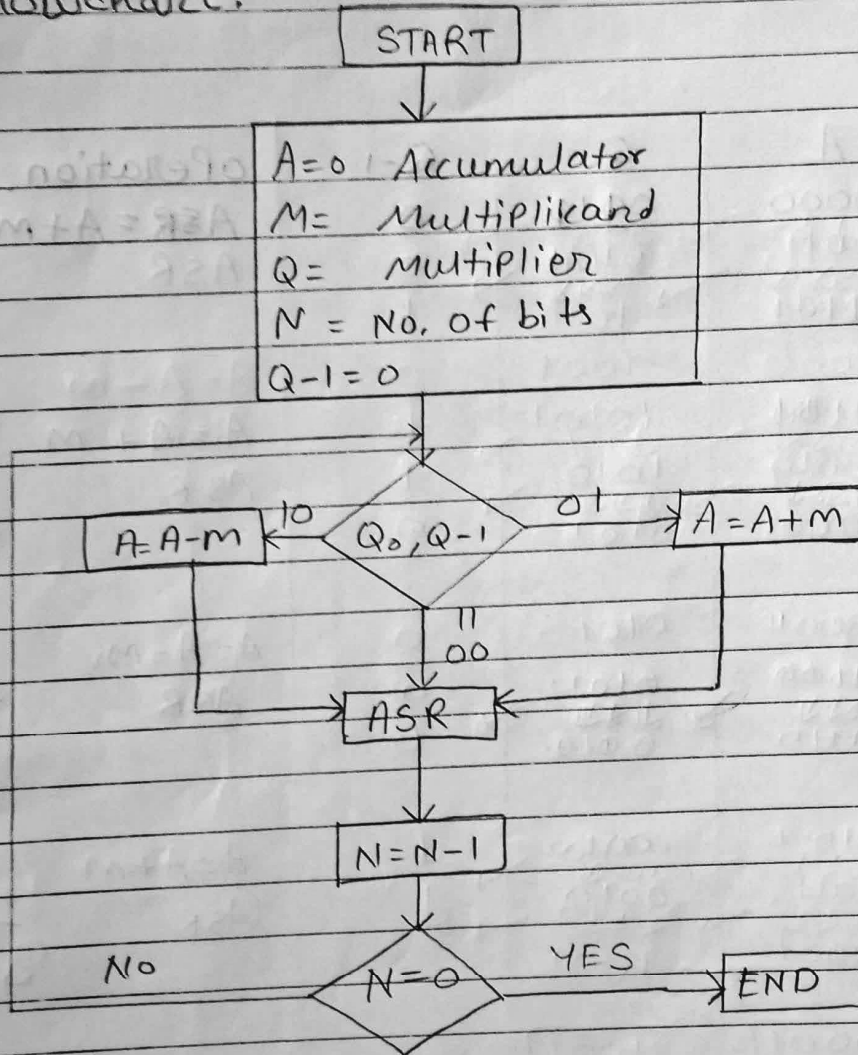
$$\Rightarrow \boxed{0001 \ 0101} \Rightarrow 3 * 7 = 21$$



Q.8 (ii) Solve 5×7 using Booth's Algorithm? (For Roll NO. 26 to 50)

Solⁿ:-

Flowchart:-





Solve: $5 \times 7 =$

$m = 5 \rightarrow 0101$

$Q = 7 \rightarrow 0111$

$-m: (1^s) \rightarrow 1010 \quad (2^s)$

$+ \quad 1$

$-m \rightarrow 1011$

	A	Q	Q-1	operation
Step:1	0000 0101 ↓ ↓ ↓ 1101	0111 0101 ↓ ↓ ↓ 1010	0 0 1	$A \oplus M + 0000$ ASR 1011
Step:2	0000 1101 ↓ ↓ ↓ 0001	0001 1010 ↓ ↓ ↓ 0101	0 1 1 0	$A = A - M$ $A = A + M$ ASR 0101 discarded
Step:3	0001 1100 ↓ ↓ ↓ 1110	0101 0101 ↓ ↓ ↓ 0010	0 0 1	$A = A - M$ ASR 0001 1100
Step:4	1110 0011 ↓ ↓ ↓ 0001	0010 0010 ↓ ↓ ↓ 1001	1 1 1	$A = A + M$ ASR 1110 0011 discarded
	0001	1001		

$5 \times 7 = 35 \quad (0010 \ 0011)$

(iii) 3rd Gen computers: (1965-1971): The third generation (1965-1971) used Integrated circuits (IC) instead of transistor. A signal IC can pack huge no. of transistor which increased the power of a computer and reduced the cost.